SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I CHRISTOPHER *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1)

Security Attacks, Threats, Risks and Security Goals (Concept Map)

Security Attacks

Passive Attacks
- Eavesdropping.
- Traffic analysis.
- Monitoring.

Active Attacks
- Modification.
- Replay.
- Masquerade.
- Denial of service.

Threats
- Malware (viruses, worms)
- Phishing / social engineering.
- Inside threats
- Natural disasters.
- Human error, misconfiguration.

Risks
- Data loss / data breach.
- Financial loss.
- Reputational damage.
- Service disruption.
- Legal / compliance issues.

INFORMATION SECURITY

Relationships
- Threats exploit vulnerabilities.
- Attacks cause harm.
- Risks represent potential losses.
- Security goals help reduce risks.

Security Goals (CIA Triad)

Confidentiality (prevents unauthorized access)

Integrity (prevents unauthorized modification)

Availability (ensures services are accessible).

Real-world example

- A phishing email steals online bank banking credentials, leading to unautho-rized access & possible financial loss.

Mitigation: Multi-factor authentication, encryption.

2)

Substitution Techniques (Concept Map)

SUBSTITUTION TECHNIQUES.

Caesar cipher
- Shift each letter by a fixed number.
- key: shift value (e.g., 3)
- Eg:- HELLO → KHOOR
- Weak against brute force attacks.

Monoalphabetic cipher
- Replaces each letter using a fixed substitution alphabet.
- key:- 26 letters.
- Eg:- A → Q, B → W, ...
- Vulnerable to frequency.

Vigenere cipher
- uses a repeating keyword to shift letters by different amounts.
- key :- KEY.
- Eg :- HELLO → RIJVS.
- Stronger than simple substitution.

Hill cipher.
- Uses a matrix multiplication & arithmetic.
- key :- key matrix.
- Eg:- Encrypts blocks of letters.
- Stronger transformation.

Comparsion :-

Key usage	Shift value	Substitution alphabet	keyword	key matrix
Complexity	low	Moderate	Moderate	Higher
Security strength	weak	weak	Stronger	Stronger.

Key Takeaway :-
- All are substitution techniques but differ in key usage and complexity.
- Hill & vigenere ciphers are generally stronger than caesar, but none are secure for modern sensitive communication.

3)

Classical Encryption Schemes (Concept Map)

Classical Encryption Schemes

Substitution Techniques
(Replace letters / symbols, keep position)

Examples:-
- Caesar cipher.
- Monoalphabetic cipher.
- Vigenere cipher.
- Hill cipher.

Transposition Techniques

Examples:-
- Rail Fence cipher.
- Columnar Transposition cipher.

Differences
- Substitution changes the symbols, transposition changes the order.
- Substitution may preserve some letter-frequency patterns.
- Transposition preserves the original letter frequencies.

Similarities
- Both uses key & encryption/decryption processes
- Both aim to provide confidentiality.
- Both are vulnerable to crypt-analysis when used alone.

Real-world Illustration.
- Caesar cipher:-
  HELLO → KHOOR
  (shift = 3).
- Transposition cipher:-
  HELLO → HLOEL
  (rearranged order).

4)

Steganography Concepts
(Concept map)

Steganography.

Hiding information in ordinary files.

Data hiding
- Conceals secret information inside another file / medium.
- Techniques:- LSB, Spatial-domain, transform-domain.
- Example:- Hiding a text message in an image / audio.

Watermarking
- Embeds identifying information into digital content.
- Types:- Visible, invisible, robust, fragile.
- Example:- Adding a digital watermark to a photograph.

Survivability
- Measures whether hidden information remains detectable after processing/modification.
- Factors:- Robustness, capacity, security.
- Example:- Robust watermark.

Relationships :-
- Data hiding conceals a message.
- Watermarking embeds identifying information.
- Survivability determines how well the data changes.

Conclusion:- Steganography support secure communication by hiding information within the ordinary digital files, while watermarking & Survivability helps to protect & verification.